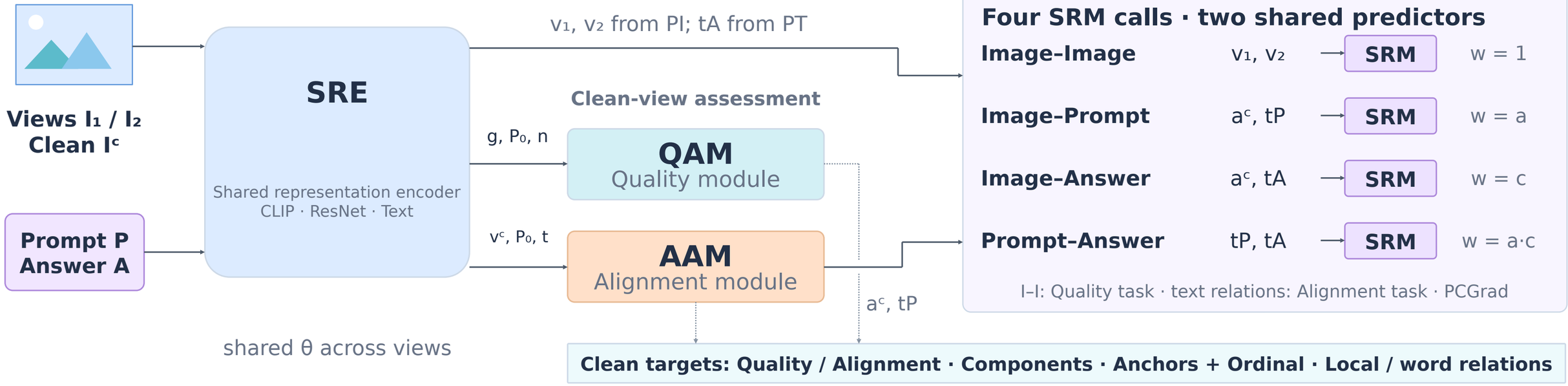
Projection-only multimodal assessment

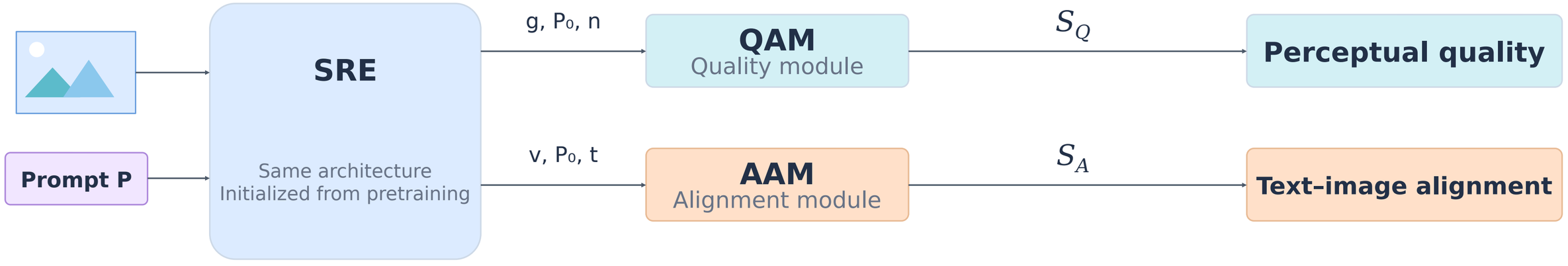
Shared representations, separate Quality and Alignment modules; no cross-attention fusion.

(a) Quality-aware multimodal pretraining



Representation initialization only · downstream scoring heads / matching layers / score gates reset

(b) Downstream quality and alignment assessment



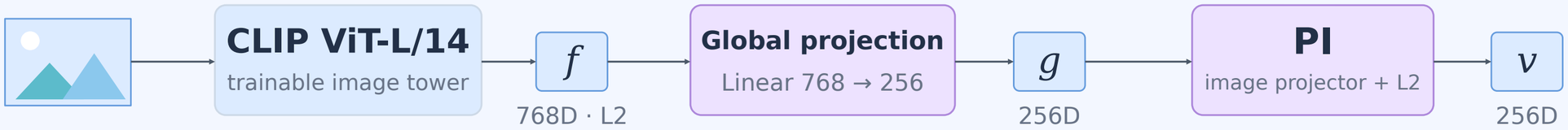
Forward data · Training supervision · Parameter initialization · ● feature fork · reuse = same tensor

projection_only_v1 · 5 trainable context tokens · frozen CLIP text weights · quality text-bank CE OFF · no negative push

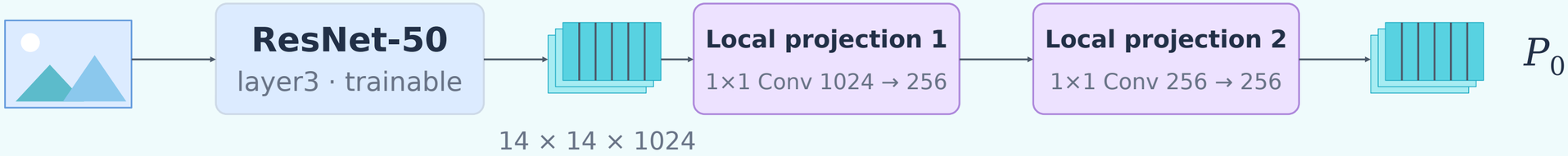
SRE | Shared Representation Encoder

One image produces independent global and local features; text uses five trainable context tokens.

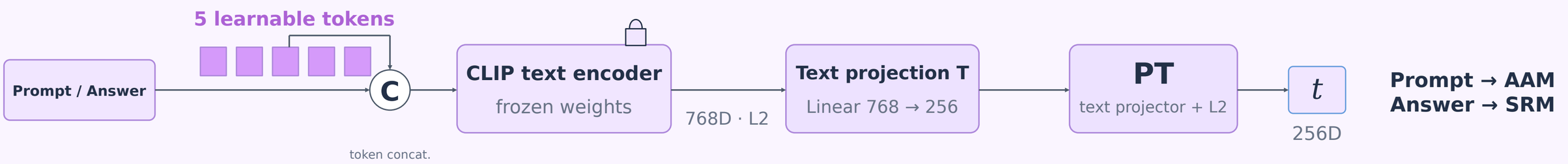
Global visual path



Independent local path



Text path



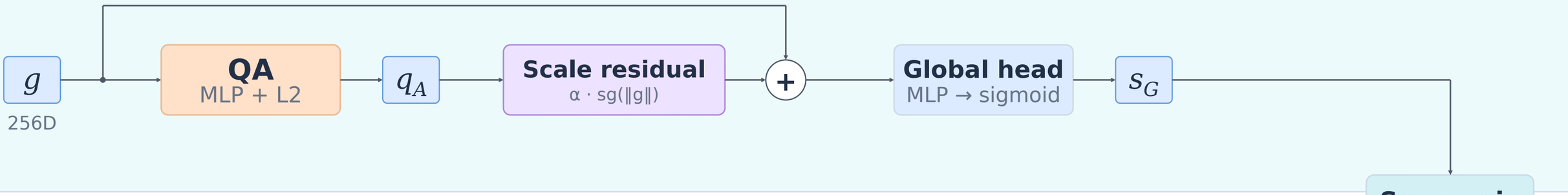


PI/PT: separate 256 → 512 → 512 → 256 MLPs. Shared SRE weights across calls; g goes directly to Quality and P_0 bypasses PI.

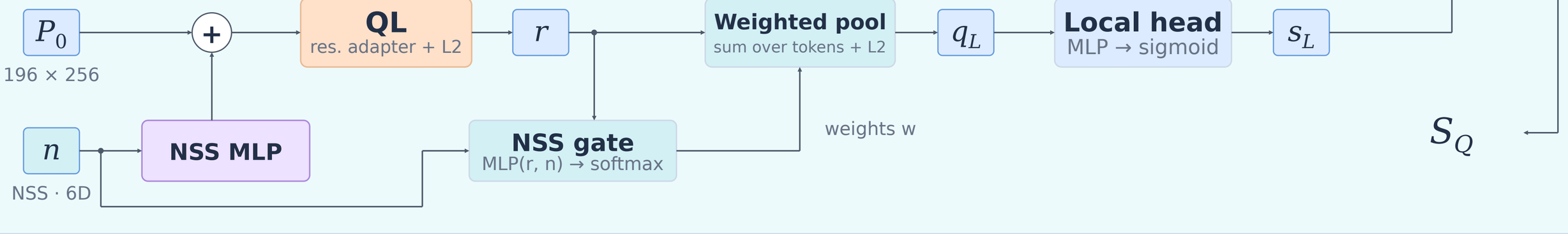
QAM | Quality Assessment Module

Global residual scoring and NSS-guided local scoring remain separate until scalar score mixing.

Global quality



Local quality



Pretraining only · reuse qA and qL above

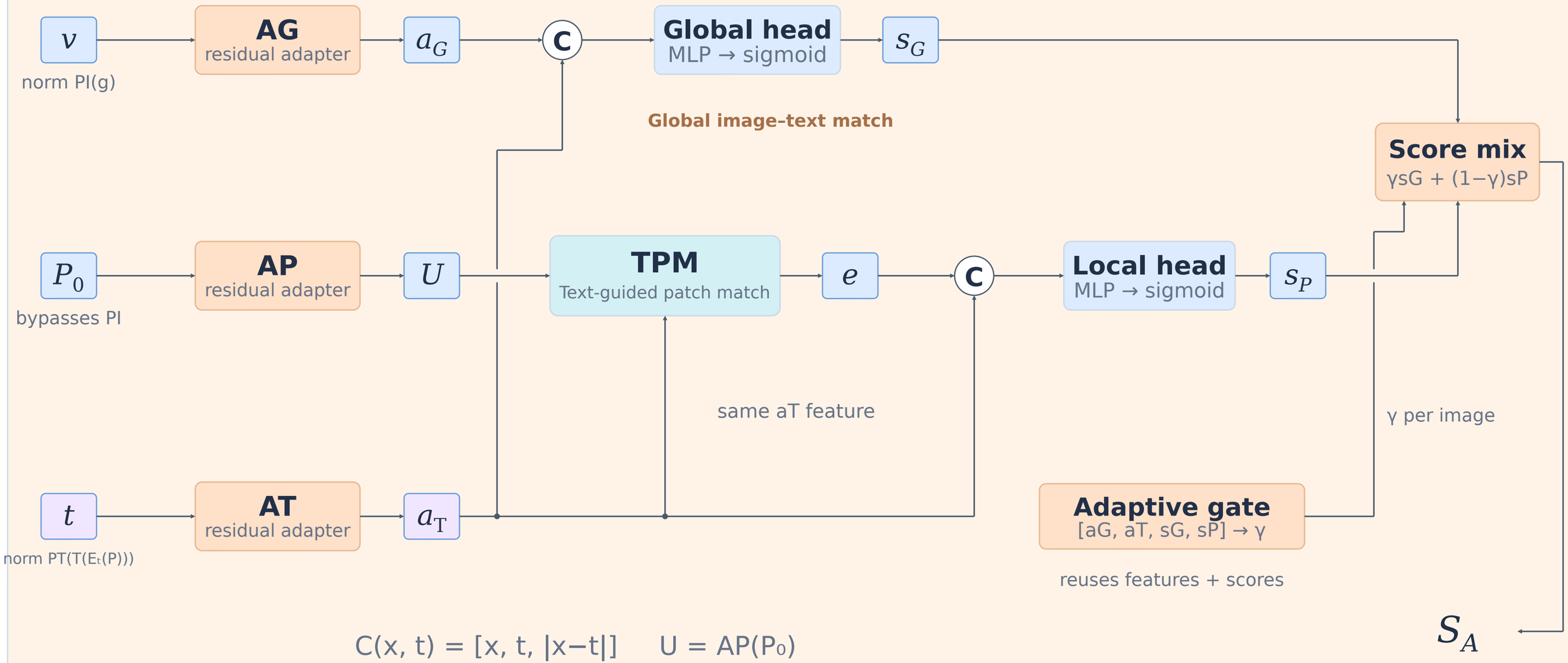


β and α are learned scalars. “+” adds features; score mix combines scalar predictions. Quality text-bank CE stays disabled.

AAM | Alignment Assessment Module

Global matching and text-guided patch evidence produce two scores, combined by an adaptive gate.

Alignment computation

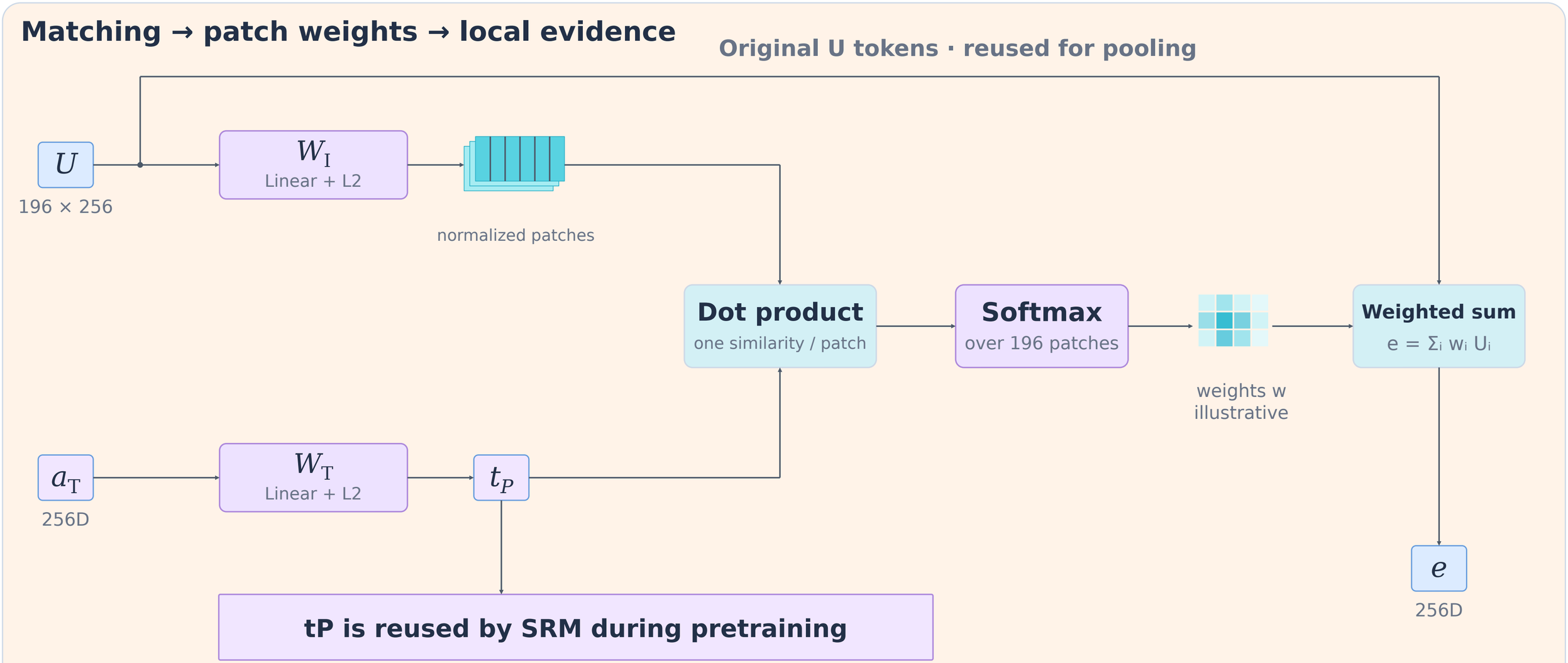


Pretraining taps: aᶜ = norm(aG); tP = norm(W_T(aT)). Answer tA comes directly from SRE, bypassing AAM.

AG / AP / AT are separate adapters. The adaptive gate reuses the four labeled inputs; TPM performs dot-product pooling, not cross-attention.

TPM | Text-guided Patch Matching

Use text to weight patches; pool the original adapted patch tokens to obtain local evidence.

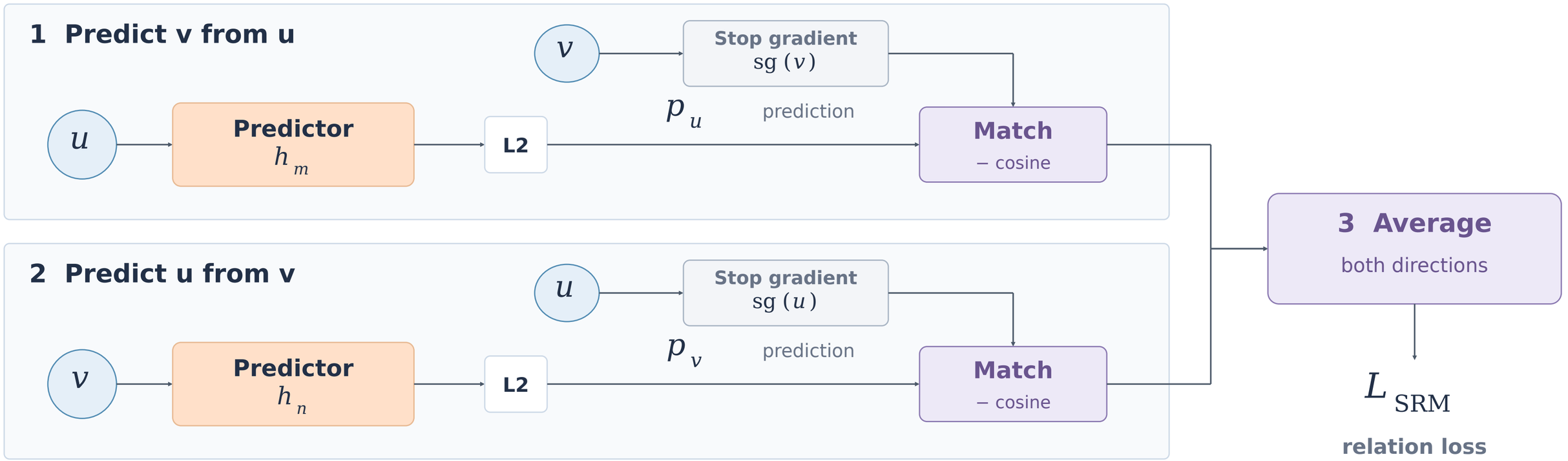


The matching projections produce attention weights only. The weighted values are U = AP(P_0).
This is scalar dot-product patch pooling, with no multi-head cross-attention or global–local fusion output.

W_I and W_T are learned Linear matching layers; both reset for downstream training. Heatmap cells are an illustrative vector schematic.

SRM | Symmetric feature matching

Predict the other feature → swap roles → average the two losses.



Same inputs, opposite roles. Only the prediction path receives gradients.

USE THE SAME MODULE FOR FOUR RELATIONS

Image ↔ Image		Image ↔ Prompt		Image ↔ Answer		Prompt ↔ Answer	
Input pair	Predictors	Input pair	Predictors	Input pair	Predictors	Input pair	Predictors
(v_1, v_2)	$(h_{\mathrm{I}}, h_{\mathrm{I}})$	(a^c, t_P)	$(h_{\mathrm{I}}, h_{\mathrm{T}})$	(a^c, t_A)	$(h_{\mathrm{I}}, h_{\mathrm{T}})$	(t_P, t_A)	$(h_{\mathrm{T}}, h_{\mathrm{T}})$

Only two predictor networks: one for images and one for text, reused across all four relations.

Current mode: positive matching only · 256-D inputs · PI/PT are outside SRM · Targets are detached, not permanently frozen